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A multiclass classification using one-versus-all approach with the differential partition sampling ensemble

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ABSTRACT

The One-versus-all(OVA) approach is one of the mainstream decomposition methods by which multiple binary classifiers are used to solve multiclass classification tasks. However, it exists the problems of serious class imbalance. This paper proposes a differential partition sampling ensemble method(DPSE) in the OVA framework. The number of majority samples and that of the minority samples in each binary training dataset are used as the upper and lower limits of the sampling interval respectively. Within this range, the construction process of the arithmetic sequence is simulated to generate the set containing multiple different sampling numbers with equal intervals. All samples are divided into safe examples, borderline examples, rare examples, and outliers according to the neighborhood information, then Random undersampling for safe samples(s-Random undersampling) and SMOTE for borderline examples and rare examples (br-SMOTE) are proposed based on the distribution characteristics of the classes. In each iteration, according to the number of differential sampling, the two methods are used to undersample or oversample the majority and minority in each binary training dataset to balance the number of positive and negative samples, which preserves the characteristic of the class structure as much as possible. Balanced training sets are used to train the binary classification model with multiple sub classifiers. The thorough experiments performed on 27 KEEL public multiclass datasets show that DPSE outperforms the typical methods in the OVA scheme, the One-versus-One scheme or direct way in classification performance.

1. Introduction

Classification is one of the important problems in the field of data mining. It is widely used in various practical fields, such as sentiment classification (Catal and Nangir, 2017), fraud detection (Nami and Shajari, 2018; Triepels et al., 2018), and fault diagnosis (Islam and Kim, 2018). Usually, the binary classification problem involves two classes, and in the multiclass classification problem, the number of classes is greater than two. The multiclass classification problem is more complicated than the binary classification problem. The methods for dealing with multiclass classification problem are mainly divided into two groups. One is to expand the binary classifier into multiclass classifier through some strategies, and it includes some typical algorithms, such as support vector machine (de Lima et al., 2018), decision tree (Guan et al., 2017), Oblique decision tree ensemble (Zhang and Suganthan, 2014; Katuwal et al., 2020), XGBOOST (Chen and Guestrin, 2016) and deep neural network (Hosaka, 2019). The second is to divide the multiclass classification problem into multiple binary problems (binarization) (Zhou et al., 2017; Liu et al., 2017). The former directly uses one classifier to deal with multiclass classification tasks. It is easier to establish a classifier to distinguish two classes than to distinguish multiple classes, and the decision boundaries of two classes may be simpler than the multiclass decision boundaries (Krawczyk et al., 2018). In contrast, decomposing the original multiclass problem into several binary sub-problems is much easier, so the second method has attracted extensive attention in practical research (Zhang et al., 2016, 2018; Li et al., 2020).

Two most popular approaches regarding binarization are one-versus-one (OVO) and one-versus-all (OVA) (Galar et al., 2011). The OVO approach divides a multiclass problem with m classes into $m(m+1)/2$ binary sub-problems, and each classifier in the OVO discriminates between a pair of classes $\{c_i, c_j\}$. When a test pattern is classified by this scheme, the score matrix R will be obtained by all the binary classifiers. Since each classifier only has the ability to classify the two classes it contains, when the instance' s true label does not belong to both classes, the classifier will give invalid discrimination (Galar et al., 2013, 2015; Zhou and Fujita, 2017). The OVA approach divides a multiclass problem with m classes into m binary sub-problems, and each classifier treats one of the classes as the positive class, and all the

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other classes as the negative class. Compared with the OVO scheme, when the dataset contains more classes, the OVA approach deploys fewer resources or uses fewer classifiers (Zhou and Fujita, 2017; Sen et al., 2016). As the number of classifiers decreases, the outputs of all binary classifiers are more simple to aggregate. Additionally, there is no invalid classifier phenomenon in the framework. Each classifier only considers a certain class and all the other classes, which simplifies the problem. However, even if there is no class-imbalance among the classes, as the number of all the other classes is larger than that of the target class, it will cause the imbalance of positive and negative sample numbers, which affects the classification effect of each binary classifier (Sen et al., 2016; Zhang et al., 2016; Li et al., 2020). How to solve the imbalance caused by OVA will be of great significance to improve the classification effect.

In recent years, binary class imbalance problems have received widespread attention (Lin et al., 2017; Douzas et al., 2018; Collell et al., 2018). In fact, in the OVA framework, each sub-problem can be regarded as a binary classification problem. Solving the imbalanced data classification problem in the OVA framework is actually solving the binary class imbalance problems. For example, Li et al. (2020) uses the OVO decomposition strategy to divide multiclass classification problems into multiple binary classification problems and then uses the oversampling with spectral clustering to balance the data of each binary class imbalance problem to effectively reduce the impact of imbalance. Zhang et al. (2016) explores the effectiveness of binary class imbalanced learning methods UnderBagging (Barandela et al., 2003), SMOTEBagging (Díez-Pastor et al., 2015), RUSBoost (Seiffert et al., 2010), SMOTEBoost (Chawla et al., 2003), SMOTE + AdaBoost (Liu et al., 2009), and EasyEnsemble (Liu et al., 2009) to solve multiclass imbalanced classification problems in the OVO framework. Experimental research proves that combining decomposition strategy and ensemble learning can improve mining imbalanced multi-class problems. Compared with the single sampling method, ensemble learning combined with data preprocessing can not only balance the number of samples but also improve diversity (García et al., 2018). Therefore, our research focuses on using ensemble learning methods to solve the class imbalance problem in the OVA framework to improve the classification performance of OVA.

In this paper, we propose a multi-classification method for multiclass imbalanced data classification by combining the OVA decomposition strategy with ensemble learning. This method proposes the differential partition sampling ensemble (DPSE) to construct a binary classification model for each binary classification problem, which can overcome the problem of data imbalance caused by the OVA decomposition strategy. DPSE combining ensemble learning and sampling methods establishes the differentiated training datasets to increase the diversity of each binary classification model. Before iteration, unlike DTE-SBD in Sun et al. (2018), DPSE calculates the differential set which includes multiple gradually increasing sampling numbers by simulating the construction process of the arithmetic sequence. In order to improve the performance of the single sampling method, all samples are divided into safe examples, borderline examples, rare examples, and outliers according to the neighborhood information before data processing. Then Random undersampling for safe samples(s-Random undersampling) and SMOTE for borderline examples and rare examples (br-SMOTE) methods that consider the distribution characteristics of the classes are proposed to balance the number of minority samples and majority samples in each iteration.

To verify the effectiveness of the proposed method, the average of each class accuracy ($MAvA$) (García et al., 2018) was used as the performance measure. 27 multiclass datasets from the KEEL dataset repository (Triguero et al., 2017) and three different base classifiers including CART (Gordon et al., 1984), Random forest (Liaw and Wiener, 2002), and SVM (Vapnic, 1998) were selected for thorough experimental research. In the OVA scheme, the differential partition sampling ensemble was compared with the typical imbalanced learning methods. Moreover, the proposed method was compared with three typical methods for solving multiclass imbalanced classification problems. The difference between the proposed method and other methods was also verified by the Friedman test and Holm–Bonferroni test (García et al., 2010).

The rest of this paper is organized as follows. We first describe the imbalanced learning problem and the solution in Section 2. Then, the approach proposed in this paper is introduced in Section 3. In Section 4, the experimental framework is presented in detail, as well as the results and discussion. Finally, the conclusions are given in Section 5.

2. Related work

In order to solve the data imbalance problem in the OVA scheme, we first introduce and analyze the existing imbalanced learning methods in Section 2.1. And then the OVA decomposition strategy is introduced in detail in Section 2.2.

2.1. Imbalanced learning method

Class-imbalance means that the number of instances in some classes is more than that of other classes in the classification problem (Roy et al., 2018). This phenomenon is widespread in real life, such as fraud detection (Nami and Shajari, 2018; Triepels et al., 2018) and biomedical diagnosis (Tan et al., 2018). In the binary problem, the class containing a large number of samples is called the majority class, and the class containing fewer instances is called the minority class. When there is a class imbalance problem in the dataset, the classifier is often more sensitive to the majority class, and reducing the recognition rate of the minority class. In recent years, in order to solve the imbalanced classification problem, many methods have been proposed (Haixiang et al., 2017). These methods are mainly divided into four groups: data level, algorithm level, cost-sensitive learning and ensemble learning (Zhang et al., 2016).

The data-level method rebalances the class distribution by sampling the data space, specifically undersampling samples of the majority class or oversampling samples of the minority class. The undersampling method mainly includes random undersampling and clustering-based undersampling (Lin et al., 2017). Random undersampling selects some samples from the majority class randomly and forms a new training set with all samples in the minority class. This method generates balanced datasets, but it has the disadvantage of strong randomness and lack of consideration of the data distribution characteristics, which will result in the loss of useful information of the majority class. The clustering-based undersampling gathers samples of the majority class into k clusters, where k is equal to the number of data samples in the minority class, and the cluster center or the data closest to the cluster center is selected as the training sample. But when the number of samples in the minority class is large, directly setting the number of samples in the minority class as the number of clusters may affect the clustering process, thereby affecting subsequent sampling. SMOTE (Chawla et al., 2002) is the most classic method in oversampling, due to its simplicity in the design of the procedure, as well as its robustness when applied to different type of problems (Fernandez et al., 2018). It randomly selects samples in the minority class to find the nearest k minority neighbors of these data and synthesizes a new few samples between these neighbors and the selected sample. Although this method can solve the imbalance between classes, it ignores the imbalance within classes, and randomly selecting is easy to bring noise data. Borderline-SMOTE1 only uses a small number of samples located in the boundary to synthesize new samples, and besides the boundary points, Borderline-SMOTE2 allows some majority samples in k neighbors to exist (Han et al., 2005). These two methods can reduce the generation of noise to a certain extent, and Sáez et al. (2016) also indicated that using information about the class structure to oversample concrete types of examples may lead to a significant improvement in classification performance. The heuristic oversampling method based on k-means and SMOTE clusters all data using k-means to focus the minority samples generation on crucial areas of the input space (Douzas et al., 2018). This method can solve not only the problem of imbalance between classes but also the problem of imbalance within classes.

The algorithm level method directly modifies the existing method or proposes a new method to solve the class imbalance problem. Collell et al. (2018) proposed a method combining the threshold movement and bagging ensemble to solve the binary imbalanced classification and multiclass imbalanced classification, and the classification accuracy depends on the threshold setting. A random forests quantile classifier for class imbalanced data was proposed in O'Brien and Ishwaran (2019). In addition, using one class learning methods (Krawczyk and Woźniak, 2016; Krawczyk et al., 2018) is also one of the ways to solve the problem of imbalanced data classification.

Cost-sensitive learning improves the recognition rate of the minority class samples by giving a higher number of misclassification costs to the minority class, for example, CS-TREE (Ling et al., 2006), CS-BPNN (Zhou and Liu, 2006). How to determine the appropriate cost is a problem that needs to be considered and is difficult to solve.

The ensemble learning method combines data-level method with ensemble learning Bagging or Boosting method. These methods have attracted a lot of attention, such as SMOTEBoost (Chawla et al., 2003), RUSBoost (Seiffert et al., 2010), Bagging-RB (Díez-Pastor et al., 2015) and DTE-SBD (Sun et al., 2018). Both SMOTEBoost and RUSBoost combine boosting algorithm with a single sampling technique. In data processing, SMOTEBoost uses SMOTE to synthesize new samples, and RUSBoost randomly removes some of the majority samples. Bagging-RB is a bagging ensemble method with random sampling number, and DTE-SBD is a Decision tree ensemble method based on SMOTE and bagging with differentiated sampling rates. This approach not only reduces the degree of data imbalance through data preprocessing but also improves classification performance by combining multiple classifiers. Galar et al. (2012) also pointed out that although the Bagging is simple, it will have a positive effect on the class-imbalance classification problem if it is reasonably combined with data preprocessing.

In multiclass datasets, because there may be more than one minority class, the boundary is more complicated, which makes it more difficult to solve the problem of class imbalance (Sáez et al., 2016). If the method of dealing with binary imbalanced classification is directly adopted, the effect may be poor or invalid (Zhang et al., 2016). And using multiple classifiers or dividing multiclass classification problems into multiple binary sub-problems can effectively solve multiclass imbalanced problems. Zhang et al. (2016) and Fernández et al. (2013) applied some methods for classifying binary imbalanced datasets to the OVO framework. The effectiveness of the combination to deal with multiclass imbalanced problems is verified through experiments. Multiple classifiers were trained in García et al. (2018) and Fernández et al. (2017). The quality of multiclass imbalanced datasets was improved by feature and instance selection in Fernández et al. (2017). The classification performance of multiclass imbalanced data was improved by dynamically selecting the appropriate classifiers for the test instance in García et al. (2018).

2.2. OVA approach

In the OVA framework, a total of m binary classification models are established for m sub-problems. When a test sample needs to be classified, all binary classification models predict its' class. A confidence for class i is $r_i (r_i \in [0,1])$ by a certain binary classification model C_i, and the output results of all binary classifiers are summarized to obtain a score vector $R = (r_1, r_2, ..r_i, \dots, r_m)$ (Zhang et al., 2013; Hong et al., 2008). There are two aggregation methods to infer the final class for OVA. MAX (Zhang et al., 2013) is a simple and effective aggregation method used in the aggregation stage of this article. When all the binary classifiers test the pattern to obtain a score vector R, the method selects the positive class corresponding to the largest element of R as the output class, that is, $Class = arg\,max_{i=1,\dots,m} r_i$. The dynamically ordered one-versus-all (DOO) method (Hong et al., 2008) does not rely on the confidence given by binary classifiers. It trains a Naive Bayes classifier on the initial dataset in advance, and it gives a test sequence of all the binary classifiers for each pattern. Each binary classifier discriminates the instance in order until a positive answer is obtained. Otherwise, the class with the largest confidence is output.

3. The proposed method: the differential partition sampling ensemble in the OVA

The total number of original training samples corresponding to each classifier is the same in the OVA framework. Considering the effectiveness of the ensemble learning with preprocessing techniques in binary imbalance learning, the method is applied to each binary dataset to solve the data imbalance problem (Zhang et al., 2016). Undersampling and oversampling may have drawbacks when used in isolation, as a single approach is not suitable for all imbalanced datasets (Nanni et al., 2015). In fact, by combining oversampling and undersampling, the number of deletions for samples in the majority class can be reduced, and the number of synthesis for samples in the minority class can also be reduced. Therefore, the proposed method mainly includes two key components: the generation of binary training datasets, the construction and aggregation of binary classification models.

3.1. The generation of binary training datasets

This section describes the generation process of binary training datasets, which mainly includes the setting of sampling numbers and the execution of the sampling method. The process about the setting of sampling numbers is described in Section 3.1.1, and the sampling methods are introduced in detail in Section 3.1.2.

3.1.1. The setting of sampling numbers

The OVA approach divides the original dataset with m classes into m binary datasets $D_i (i = 1, 2, ..m)$, where the positive class in each dataset corresponds to one of the classes, and the negative class is composed of all the other classes. The imbalance ratio of each binary dataset is different, and the diverse training sets are of great benefit to improve the classification effect of the classification model (Sun et al., 2018). In order to generate multiple training sets whose number of samples is gradually increasing, the number of sampling is set for each binary dataset based on the idea of incremental arithmetic progression. The idea of the sampling is similar to bootstrap aggregation. The original training set is sampled several times by increasing the number of samples regularly in order to produce multiple balanced training sets with different distributions.

Fig. 1 represents the generation process of multiple balanced training sets in a binary classification problem. The ith balanced binary training set is used in the ith iteration. In this way, we do not need to consider the problem that the imbalance ratio corresponding to each binary dataset is different. In addition, the difference in the number of samples may make the sample subsets different, which will increase the diversity of the model. The number of sampling is calculated as follows:

(1) The interval of the number of samples $[count(less_i), count(more_i)]$ is determined where $count(less_i)$ is the total number of minority samples in the ith binary training dataset and $count(more_i)$ is the total number of majority samples;
(2) The set of the number of sampling $num_i = \{num_{i1}, num_{i2}, \dots, num_{iT}\}$ is obtained according to formula (1) where T is the total number of classifiers (the number of training set generation) and num_{it} is the number of sampling for the tth classifier corresponding to the ith binary dataset. If num_{it} is a decimal, only the integer part is replaced the value. num_{i1} is the lower limit $count(less_i)$ of the interval, and num_{iT} is the upper limit $count(more_i)$ of the interval.

$$num_{it} = \frac{(count(more_i) - count(less_i))}{T-1} * (t-1) + count(less_i) \qquad (1)$$

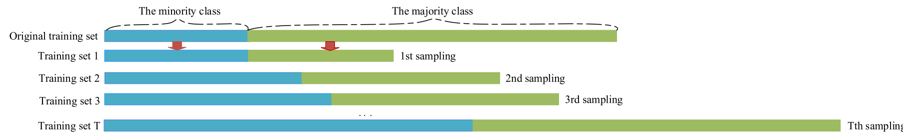

Fig. 1. The process of training set generation.

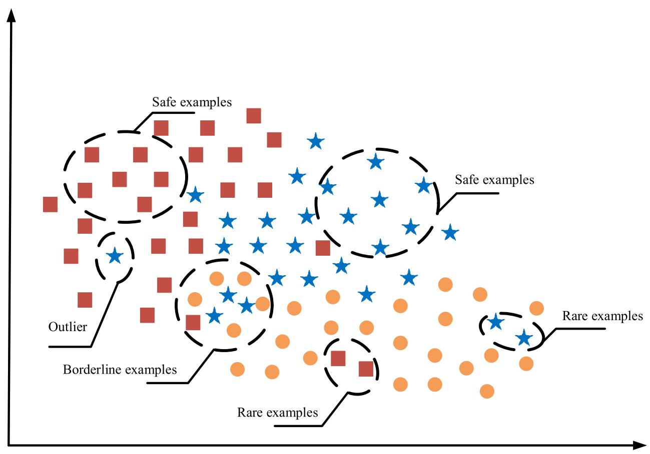

Fig. 2. The distribution of different samples.

3.1.2. s-Random undersampling and br-SMOTE

The sample located in different regions is related to its' importance, and safe examples are easier to be identified than borderline examples, and outliers tend to affect the decision of the classifier (García et al., 2018). We first determine the type of training samples by analyzing the neighborhood samples' class which includes safe examples, borderline examples, rare examples, and outliers, as shown in Fig. 2. Then, all the outliers are removed. The specific analysis process is as follows:

(1) Firstly, the k samples closest to the example in the training set are calculated by the HVDM distance metric (Wilson and Martinez, 1997) and regarded as its neighborhood samples. The value k is used to analyze the neighborhoods of an imbalanced dataset($k = 5$ in this paper). There is an instance x whose actual label is y.

(2) Then, the relationship between the classes of samples in the neighborhood and the class y of x is analyzed to determine the type of x. The example x is considered to be safe, if, at least, 4 of the 5 nearest neighbors share the same class with x. And x is considered as borderline one if 2 or 3 of its 5 nearest neighbors share its label. If only one of the 5 nearest neighbors is the same class of x and its neighbor sample contains at most one sample of the same class, x is considered as a rare example. The example x is considered to be the outlier if it is surrounded by examples from different classes (Sáez et al., 2016).

Next, according to the number of sampling, the majority samples are undersampled and the minority samples are oversampled for each binary dataset. In the undersampling process, considering the dense distribution of safe points, all borderline examples maj_{ib} and rare points maj_{ir} of the majority are reserved, and only the safe examples maj_{is} of the majority are randomly undersampled. The number of samples $count(maj'_{is})$ can be calculated according to formula (2). If the number of maj_{is} is less than that of other types of samples or the number of other types is greater than the number of sampling, random undersampling is performed on all the majority. The details are shown as the Algorithm 1. In oversampling, all safe examples of the minority minis are retained. Borderline examples and rare examples of the minority are used to synthesize new samples by SMOTE, because they tend to have a decisive effect on the classification result (García et al., 2018). The number of samples $count(min'_{ir}) + count(min'_{ib})$ can be calculated according to formula (3). If there is no borderline example and rare example, all the minority are oversampled by SMOTE. The specific details are shown in Algorithm 2.

$$count(maj'_{is}) = num_{it} - count(maj_{ib}) - count(maj_{ir}) \tag{2}$$

$$count(min'_{ir} + min'_{ib}) = num_{it} - count(min_{is}) \tag{3}$$

Algorithm 1 s-Random undersampling

Input: D_{maj} marked as 1, 2, 3; 1 represents the safe example,and 2 represents the borderline example, and 3 represents the rare example; *num* is the number of sampling;
Output: D'_{maj} obtained after sampling;
Count the number of samples of type 1 $count(Safe)$ in D_{maj}, and the total number of samples of other types $count(!Safe)$.
if $count(!Safe) > count(Safe)$ or $count(!Safe) > num$ **then**
  Select some samples randomly from D_{maj}, and get a subset D'_{maj}, and satisfy the number of samples it contains equal to *num*
else
  Retain samples of type 2 and 3, and randomly select some samples from the sample of type 1, and get a subset D'_{majs}, satisfying the number of samples it contains equal to $num - count(!Safe)$, let $D'_{maj} = D'_{majs} \cup D_{majb} \cup D_{majr}$
end if

Algorithm 2 br-SMOTE

Input: D_{min} marked as 1, 2, 3; 1 represents the safe example,and 2 represents the borderline example, and 3 represents the rare example; *num* is the number of sampling;
Output: D'_{min} obtained after sampling;
Count the number of samples of type 1 $count(Safe)$ in D_{min}, and the total number of samples of other types $count(!Safe)$.
if $count(!Safe) == 0$ **then**
  Oversampling D_{min} by SMOTE, and get a subset D'_{min}, and satisfy the number of samples it contains equal to *num*
else
  Retain samples of type 1, and oversampling samples of type 2 and 3 using SMOTE, and get a subset D'_{minbr}, satisfying the number of samples it contains equal to $num - count(!Safe)$, let $D'_{min} = D_{mins} \cup D'_{minbr}$
end if

3.2. The construction and aggregation of binary classification models

The differential partition sampling ensemble is proposed to solve the data imbalance problem in the OVA scheme and build binary classification models. First, all training samples are divided into safe examples, borderline examples, rare examples, and outliers according to the class information of neighborhood samples, and all outliers are deleted. The specific division process is described in Section 3.1.2. Then, the original dataset D with m classes is divided into m binary datasets D_i, and the set of the number of sampling $num_i = \{num_{i1}, num_{i2}, \ldots, num_{iT}\}$ for each binary dataset in the range of $[count(less_i), count(more_i)]$ is calculated according to formula (1). The calculation process is described in Section 3.1.1. Secondly, based on the Bagging, the binary classification model with T sub-classifiers (total number of training set generation) is established for each binary classification problem. In the tth iteration, according to the number of sampling num_{it}, the number of the majority samples is reduced and that of the minority samples is increased to form a balanced subset for training a binary classifier. And br-SMOTE is used to oversample the minority and s-Random undersampling is used to undersample the majority in the process of balancing the number of samples. The specific processes of br-SMOTE and s-Random undersampling are shown in pseudocode Algorithm 1 and Algorithm 2. If there is no imbalance between the number of positive and negative samples in the binary training subset, $T = 1$ and the original binary training set D_i is used to train a binary model. Finally, these binary classifiers are ensembled by the simple averaging method to obtain the final binary classification model. The pseudocode in Algorithm 3 describes the detailed process about binary classification models' construction.

Algorithm 3 One versus all with the differential partition sampling ensemble

Input: Original training dataset with m classes—D,the number of training set generation T;
Output: m binary classifier models—$h_i(x)$;

```
Divide all training samples into safe examples, borderline examples, rare examples, and outliers,and delete all outliers
Create m binary datasets D_i from D, where the target class is the positive class p, and all other classes are the negative class n
for i = 1 to m do
  if count(p in D_i) ≠ count(n in D_i) then
    if count(p in D_i) < count(n in D_i) then
      more = D_n in D_i;less = D_p in D_i
    else
      more = D_p in D_i;less = D_n in D_i
    end if
    With [count(less), count(more)] as the interval, and the number of sampling' set num = {num_1, , num_2, ..., num_T} is obtained by formula (1),where count(more) is the total number of samples in majority class, and count(less) is the total number of samples in minority class.
    for t = 1 to T do
      sample_num = num[i]
      Oversampling less by br-SMOTE to get a subset less', satisfying count(less') = sample_num
      Undersampling more by s-Random undersampling to get a subset more', satisfying count(more') = sample_num
      D'_i = more' ∪ less',train a classifier h_it(x) on D'_i
    end for
  else if count(p in D_i) == count(n in D_i) then
    D'_i = D_i,train a classifier h_it(x) on D'_i,T = 1
  end if
  Get the final model of each binary classifier
```

$$h_i(x) = \frac{1}{T}\sum_{t=1}^{T} h_{it}(x)$$

```
end for
```

The aggregation of all the binary models is performed with the Max strategy (Zhang et al., 2013). Each of the binary classifier $h_i(x)$ predicts the unknown instance, and the output vector $\boldsymbol{r} = \{r_1, r_2, r_3, \ldots, r_m\}$ is obtained, where r_i is the positive class' confidence of the ith binary classifier output. The maximum value of $\boldsymbol{r}$ is calculated, and the label corresponding to the value is output.

4. Experiment and evaluation

In this section, in order to verify the effectiveness of the proposed method, a comprehensive experimental analysis was performed on 27 KEEL public multiclass datasets using CART, Random forest, and SVM as the benchmark classifier respectively. The datasets chosen are described in Section 4.1, and the benchmark classifiers with their parameter settings are described in Section 4.2. The measure to evaluate the performance of the methods is presented in Section 4.3. In Section 4.4, the differential partition sampling ensemble proposed was compared with the existing typical imbalanced learning methods in the OVA scheme. In addition, in order to further prove the effectiveness of the proposed method, it was compared with typical methods for solving such problems which are in the One-versus-One scheme or in a direct way in Section 4.5.

4.1. Datasets

In this study, the experimental analysis was performed on 27 different multiclass datasets from the KEEL dataset repository. The main features of each dataset are shown in Table 1. For each dataset, the number of instances (#Exa), the number of features (#Att), the number of numeric attributes (#Num), the number of nominal attributes (#Nom), the number of classes (#cla) and imbalance ratio (#IR) are included. And IR is the ratio of the number of the majority samples and that of the minority samples. In this case, the class with the maximum number of examples is the majority, and the class with the minimum number of examples is the minority. In order to ensure the accuracy of the results, all experiments' results were obtained by means of 20 times five-fold cross-validation. That is, every four folds containing 80% of the examples of the dataset is the training set, and the test set contains 20% of the examples of the dataset.

Table 1
Summary description of the datasets used in the experiments.

Dataset	#Exa	#Att	#Num	#Nom	#cla	#IR
balance scale	625	4	4	0	3	5.80
car	1728	6	0	6	4	18.62
dermatology	358	34	33	1	6	5.55
ecoli	336	7	7	0	8	71.5
flare	1066	6	0	6	6	7.70
glass	214	9	9	0	6	8.44
page-blocks	5472	10	10	0	5	175.46
penbased	10 992	16	16	0	10	1.08
satimage	6435	36	36	0	7	2.45
shuttle	58 000	9	9	0	7	4559
segment	2310	19	19	0	7	1
vehicle	846	19	19	0	4	1.10
vowel	990	13	13	0	11	1
wine	178	13	13	0	3	1.48
yeast	1484	8	8	0	4	92.6
zoo	101	16	0	16	7	10.25
winequality-red	1599	11	11	0	6	68.1
winequality-white	4898	11	11	0	7	436.9
contraceptive	1473	9	6	3	3	1.89
hayes-roth	160	4	4	0	3	2.1
thyroid	720	21	15	6	3	39.18
newthyroid	215	5	5	0	3	5
lymphography	148	18	13	5	4	40.5
led7digit	500	7	0	7	10	1.54
splice	3190	60	0	60	3	2.16
automobile	159	25	15	10	6	16
cleveland	297	13	5	8	5	12.31

Table 2
The parameter setting for base classifier.

Classifier	Parameters
CART	$min\_samples\_split = 2$
	$min\_samples\_leaf = 1$
Random forest	$n\_estimators = 100$
	$criterion =$ "$gini$"
SVM	$C = 1$
	$kernel =$ "rbf"

4.2. Classifier and parameter setting

In this section, three different classification algorithms used as the benchmark classifier respectively in the experiment are given, namely CART, Random forest and SVM. CART is a decision tree with feature selection based on Gini index (Gordon et al., 1984). It is mainly composed of two parts: the generation of decision tree and the decision tree pruning. The decision tree is generated by dividing the input (feature) space into finite elements. Then, the optimal sub-tree is selected by pruning the generated tree with the goal of minimizing the loss function. Random forest is one of the representative algorithms in ensemble learning (Liaw and Wiener, 2002). In addition to ensemble multiple decision trees by Bagging, it also randomly chooses attributes when constructing each decision tree, and then selects features from these attributes, which further increases the difference of sub-learners to improve the generalization performance of the final ensemble model. SVM finds a separated hyperplane in the feature space. The separated hyperplane divides the output space into two parts, one containing the positive class instances and the other containing the negative class instances (Vapnic, 1998). In all experiments, the construction of the

Table 3
The $MAvA$ (%) for different imbalanced learning methods using CART as a classifier.

	OVA	SMOTE (Chawla et al., 2002)	k-means-SMOTE (Douzas et al., 2018)	Bagging-RB (Díez-Pastor et al., 2015)	DPSE
balance scale	55.4 ± 0	54.99 ± 1.68	55.22 ± 1.36	**58.24 ± 1.65**	58.1 ± 0.44
car	94.81 ± 0	84.43 ± 1.27	86.1 ± 1.4	93.93 ± 2.35	**98.3 ± 0.24**
dermatology	91.82 ± 0	89.3 ± 0.59	91.52 ± 1.13	**94.62 ± 0.94**	94.18 ± 0.43
ecoli	50.95 ± 0	48.1 ± 1.77	55.72 ± 1.57	**60.03 ± 4.39**	59.49 ± 2.32
flare	59.96 ± 0	60.28 ± 0.74	60.49 ± 0.78	60.27 ± 2.06	**62.27 ± 0.55**
glass	64.35 ± 0	58.79 ± 2.16	61.68 ± 2.78	67.38 ± 3.71	**71.22 ± 1.73**
page-blocks	80.29 ± 0	79.75 ± 0.76	80.23 ± 1.06	**89.64 ± 2.69**	89.01 ± 0.32
penbased	96.8 ± 0	94.51 ± 0.19	95.79 ± 0.19	97.83 ± 0.35	**98.68 ± 0.05**
satimage	85.93 ± 0	80.88 ± 0.28	85.38 ± 0.36	86.38 ± 1.68	**89.12 ± 0.17**
shuttle	93.8 ± 0	93.89 ± 0.03	93.82 ± 0.02	**96.84 ± 0.18**	95.88 ± 0.01
segment	96.71 ± 0	95.57 ± 0.32	97.64 ± 0.23	97.03 ± 0.38	**97.68 ± 0.1**
vehicle	71.02 ± 0	65.86 ± 1.2	66.21 ± 1.2	73.15 ± 1.21	**74.67 ± 0.39**
vowel	86.16 ± 0	85.23 ± 1.06	86.34 ± 0.97	81.82 ± 2.05	**89.62 ± 0.46**
wine	90.09 ± 0	90.65 ± 1.38	90.01 ± 1.09	**94.78 ± 1.66**	94.19 ± 0.6
yeast	52.49 ± 0	45.69 ± 0.81	49.12 ± 1.1	52.68 ± 1.55	**54.13 ± 0.67**
zoo	82.22 ± 0	82.22 ± 0	81.56 ± 1.69	88.44 ± 3.45	**92.35 ± 1.18**
winequality-red	34.25 ± 0	36.4 ± 2.43	35.5 ± 2.02	36.32 ± 2.16	**39.38 ± 0.6**
winequality-white	38.76 ± 0	38.51 ± 1.29	38.75 ± 1.38	39.01 ± 2.03	**43.04 ± 0.72**
contraceptive	45.94 ± 0	45.38 ± 0.72	45.05 ± 0.86	47.4 ± 1.4	**48.76 ± 0.6**
hayes-roth	74.5 ± 0	74.5 ± 1.64	74.54 ± 1.84	83.33 ± 2.9	**86.25 ± 1.2**
thyroid	93.98 ± 0	93.25 ± 0.34	93.09 ± 0	95.06 ± 2.38	**95.07 ± 0.83**
newthyroid	90.79 ± 0	91.32 ± 1.2	90.86 ± 0.49	92.21 ± 1.76	**94.03 ± 0.55**
lymphography	72.52 ± 0	68.24 ± 3.58	70.68 ± 3.82	75.65 ± 3.98	**83.04 ± 0.76**
led7digit	70.02 ± 0	69.94 ± 0.32	70.14 ± 0.41	66.15 ± 0.83	**70.52 ± 0.59**
splice	89.11 ± 0	89.15 ± 0.27	89.17 ± 0.31	92.28 ± 0.42	**93.86 ± 0.2**
automobile	70.13 ± 0	67.37 ± 2.28	66.61 ± 2.8	73.65 ± 3.47	**80.96 ± 0.75**
cleveland	28.07 ± 0	28.97 ± 1.95	30.1 ± 3	30.7 ± 2.2	**31.59 ± 1.63**
Average	72.61 ± 0	70.71 ± 1.12	71.90 ± 1.25	74.99 ± 1.99	**77.23 ± 0.67**
Ranking($p = 7.13e{-14}$)	3.63	4.22	3.74	2.19	1.22

Table 4
The $MAvA$ (%) for different imbalanced learning methods using Random forest as a classifier.

	OVA	SMOTE (Chawla et al., 2002)	k-means-SMOTE (Douzas et al., 2018)	Bagging-RB (Díez-Pastor et al., 2015)	DPSE
balance scale	**62.5 ± 0**	61.66 ± 0.26	61.22 ± 0.33	59.89 ± 1.77	60.35 ± 0.1
car	94.44 ± 0	93.08 ± 1.23	92.68 ± 0.81	87.67 ± 5.07	**95.64 ± 0.21**
dermatology	97.5 ± 0	97.5 ± 2.28	97.48 ± 0.07	96.65 ± 0.81	**97.59 ± 0.34**
ecoli	71.69 ± 0	64.43 ± 1.5	69.82 ± 1.54	61.15 ± 4.05	**71.77 ± 0.51**
flare	60.08 ± 0	59.69 ± 0.34	59.95 ± 0.44	58.26 ± 2.01	**60.99 ± 0.46**
glass	69.9 ± 0	62.27 ± 1.56	64.63 ± 1.46	70.63 ± 3.96	**77.61 ± 0.88**
page-blocks	84.27 ± 0	90.88 ± 0.37	89.05 ± 0.72	89.77 ± 2.59	**92.05 ± 0.25**
penbased	99.1 ± 0	**99.2 ± 0.03**	99.17 ± 0.04	98.38 ± 0.37	99.11 ± 0.07
satimage	89.08 ± 0	89.69 ± 0.13	89.56 ± 0.1	87.31 ± 1.55	**89.73 ± 0.1**
shuttle	93.42 ± 0	94.51 ± 0.67	**95.51 ± 1.28**	93.51 ± 0.84	94.93 ± 0.74
segment	97.58 ± 0	**98.22 ± 0.11**	98.11 ± 0.1	97.16 ± 0.53	98.03 ± 0.09
vehicle	75.95 ± 0	76.97 ± 0.7	76.62 ± 0.67	71.99 ± 1.32	**77.3 ± 0.27**
vowel	95.56 ± 0	95.45 ± 0.31	96.06 ± 0.8	93.94 ± 1.89	**96.09 ± 0.36**
wine	98.63 ± 0	**98.99 ± 0.18**	98.79 ± 0.22	95.79 ± 1.12	98.78 ± 0.79
yeast	**55.95 ± 0**	53.39 ± 1.4	54.59 ± 1.62	45.91 ± 1.34	54.93 ± 0.54
zoo	87.14 ± 0	**90 ± 0**	89.14 ± 2.09	88.03 ± 2.86	**90 ± 0.82**
winequality-red	35.96 ± 0	39.78 ± 0.48	40.1 ± 1.12	35.82 ± 1.85	**40.26 ± 0.41**
winequality-white	39.06 ± 0	42 ± 0.22	41.8 ± 0.31	34.69 ± 1.94	**44.67 ± 0.5**
contraceptive	50.09 ± 0	50.14 ± 0.34	50.08 ± 0.54	48.3 ± 1.36	**50.51 ± 0.13**
hayes-roth	86.18 ± 0	84.01 ± 0.54	85.02 ± 0.49	81.41 ± 3.22	**87.45 ± 0.51**
thyroid	86.51 ± 0	93.16 ± 1.23	92.59 ± 0.75	93.62 ± 2.32	**96.16 ± 0.37**
newthyroid	94.25 ± 0	95.99 ± 0.6	94.98 ± 0.75	95.54 ± 1.57	**96.54 ± 0**
lymphography	78.62 ± 0	75.74 ± 2.5	76.58 ± 2.44	72.01 ± 4.21	**81.47 ± 0.31**
led7digit	70.8 ± 0	70.66 ± 0.38	**71.21 ± 0.58**	67.87 ± 2.74	70.85 ± 0.43
splice	95.07 ± 0	94.69 ± 0.17	94.79 ± 0.26	87.66 ± 2.68	**95.58 ± 0.03**
automobile	**79.43 ± 0**	78.78 ± 0.78	77.3 ± 2.09	60.13 ± 3.59	79.35 ± 0.68
cleveland	28.5 ± 0	31.01 ± 1.74	**31.27 ± 1.22**	30.99 ± 1.82	29.87 ± 0.92
Average	76.94 ± 0	77.11 ± 0.74	77.34 ± 0.83	74.23 ± 2.2	**78.8 ± 0.4**
Ranking($p = 2.46e{-09}$)	3.24	2.78	2.89	4.48	1.61

three classifiers relied on the sklearn package whose version is 0.19.2 in the Python3 software (Nelli, 2015), and the values of different parameters used in the base classifiers are presented in Table 2.

4.3. Performance measure

Since each class in the multiclass classification problem has the same importance, the average of each class accuracy ($MAvA$) is used

Table 5
The $MAvA$ (%) for different imbalanced learning methods using SVM as a classifier.

	OVA	SMOTE (Chawla et al., 2002)	k-means-SMOTE (Douzas et al., 2018)	Bagging-RB (Díez-Pastor et al., 2015)	DPSE
balance scale	70.33 ± 0	79.58 ± 1.11	81.21 ± 1.07	76.58 ± 3.63	**81.87 ± 0.39**
car	91.83 ± 0	92.34 ± 0.8	92.66 ± 1.15	92.28 ± 2.19	**94.52 ± 0.23**
dermatology	96.85 ± 0	**97.32 ± 0.14**	97.25 ± 0.18	96.48 ± 0.66	97.15 ± 0.1
ecoli	70.56 ± 0	66.31 ± 1.08	69.25 ± 0.61	67.86 ± 2.35	**71.89 ± 0.15**
flare	58.92 ± 0	59.47 ± 0.23	59.21 ± 0.85	57.54 ± 2.26	**62.35 ± 0.41**
glass	59.86 ± 0	**64.11 ± 1.75**	61.49 ± 3.06	60.99 ± 4.73	63.49 ± 0.74
page-blocks	71.93 ± 0	75.19 ± 0.67	79.53 ± 2.25	83.32 ± 3.33	**86.87 ± 0.14**
penbased	**99.41 ± 0**	97.99 ± 0.02	99.31 ± 0.15	99.21 ± 0.11	99.37 ± 0.01
satimage	87.49 ± 0	82.85 ± 0.09	**87.89 ± 0.24**	85.20 ± 1.53	87.1 ± 0.05
shuttle	77.89 ± 0	**93.92 ± 0.06**	93.8 ± 0.04	82.65 ± 0.09	90.03 ± 0.05
segment	93.42 ± 0	90.45 ± 0.05	93.53 ± 0.23	91.8 ± 0.9	**93.78 ± 0.07**
vehicle	78.07 ± 0	70.1 ± 0.41	75.37 ± 2.08	74.79 ± 1.42	**78.15 ± 0.28**
vowel	92.12 ± 0	90.08 ± 0.27	92.57 ± 0.65	92.15 ± 2.01	**93.59 ± 0.17**
wine	97.52 ± 0	**98.56 ± 0.15**	98.35 ± 0.69	97.67 ± 0.81	97.52 ± 0
yeast	54.9 ± 0	48.21 ± 1.01	54.48 ± 0.71	55.57 ± 1.29	**57.34 ± 0.33**
zoo	91.75 ± 0	90.2 ± 1.72	**92.86 ± 1.24**	88.04 ± 1.78	91.75 ± 0
winequality-red	28.82 ± 0	36.51 ± 0.8	**37.91 ± 0.72**	32.41 ± 2.11	35.92 ± 0.38
winequality-white	25.5 ± 01	30.45 ± 0.41	31.52 ± 0.43	30.08 ± 1.87	**36.77 ± 0.66**
contraceptive	49.62 ± 0	49.99 ± 0.28	50.39 ± 0.49	43.66 ± 3.1	**56.1 ± 0.22**
hayes-roth	86.06 ± 0	58.02 ± 1.11	60.27 ± 2.5	77.55 ± 4.91	**86.56 ± 0.37**
thyroid	64.87 ± 0	61.07 ± 0.6	59.78 ± 1.12	75.52 ± 3.9	**80.52 ± 0.05**
newthyroid	94.22 ± 0	**94.39 ± 0.53**	87.76 ± 1.67	95.84 ± 2.22	94.33 ± 0.49
lymphography	74.35 ± 0	77.49 ± 2.32	**78.4 ± 2.38**	71.08 ± 5.52	76.73 ± 2.37
led7digit	72.65 ± 0	**73.81 ± 0.6**	71.97 ± 0.43	63.74 ± 3.24	71.76 ± 0.39
splice	85.31 ± 0	84.09 ± 0.19	85.31 ± 0.21	81.93 ± 3.61	**85.54 ± 0.06**
automobile	56.73 ± 0	52.59 ± 0.62	59.39 ± 1.09	52.53 ± 3.66	**59.4 ± 1.3**
cleveland	28.94 ± 0	35.27 ± 1.81	**36.23 ± 1.92**	32.18 ± 2.63	32.27 ± 1.01
Average	72.59 ± 0	72.24 ± 0.7	73.62 ± 1.04	72.54 ± 2.44	**76.39 ± 0.39**
Ranking($p = 9.71e{-}06$)	3.57	3.22	2.46	3.85	1.89

Table 6
Results of Holm–Bonferroni test for comparing DPSE with OVA, SMOTE (Chawla et al., 2002), k-means-SMOTE (Douzas et al., 2018) and Bagging-RB (Díez-Pastor et al., 2015) ($\alpha = 0.05$).

Comparison	Corrected p-value (Holm–Bonferroni)
DPSE vs. OVA(CART)	2.2e−05
DPSE vs. SMOTE(CART)	2.2e−05
DPSE vs. k-means-SMOTE(CART)	2.2e−05
DPSE vs. Bagging-RB(CART)	1.3e−04
DPSE vs. OVA(Random forest)	6.9e−04
DPSE vs. SMOTE(Random forest)	2.0e−03
DPSE vs. k-means-SMOTE(Random forest)	2.0e−03
DPSE vs. Bagging-RB(Random forest)	4.0e−05
DPSE vs. OVA(SVM)	2.7e−04
DPSE vs. SMOTE(SVM)	4.6e−03
DPSE vs. k-means-SMOTE(SVM)	3.7e−02
DPSE vs. Bagging-RB(SVM)	6.9e−05

as the performance measure metric in this paper. The accuracy of each class (ACC_i) is the ratio of the number of correct classifications in each class and the total number of such class, and m is the total number of classes. The specific definition is as follows:

$$MAvA = \frac{1}{m}\sum_{i=1}^{m} ACC_i \quad (4)$$

4.4. Comparison with typical imbalanced learning methods

In this section, in order to verify the effectiveness of the differential partition sampling ensemble (DPSE) in the OVA scheme, it was compared with SMOTE (Chawla et al., 2002) and k-means-SMOTE (Douzas et al., 2018) which are typical oversampling methods, Bagging-RB (Díez-Pastor et al., 2015) which is an ensemble method with hybrid sampling techniques, and OVA without sampling. The number of training set generation T was set to 30, the number of clusters in the k-means-SMOTE oversampling was set to 5, and the number of neighbors was set to 5. Table 3, Tables 4 and 5 respectively correspond to the $MAvA$ obtained by CART, Random forest and SVM as the base classifier respectively on different datasets. The Friedman test and Holm–Bonferroni test were used to analyze the difference of different methods.

The best results for each dataset in three tables are shown in bold. Observing the results in Tables 3, 4, and 5, it is obvious to see that the proposed method achieves higher $MAvA$ on most datasets than other methods. Whether taking CART, Random forest or SVM as the classifier, it obtains smaller standard deviations on most datasets than other imbalanced learning methods. It can also be concluded from three tables that a single oversampling technique in the OVA scheme is not very stable and the number of positive and negative samples is important to the ensemble learning method. In addition, the Friedman test results in the last row of Table 3, Tables 4 and 5 show that under the condition that $\alpha = 0.05$, that is, the confidence is 95%, the DPSE ranks first with $p < 0.05$. The statistical results of Holm–Bonferroni test in Table 6 can also indicate that the proposed method outperforms all other methods. Therefore, it can be concluded that the DPSE is more effective in solving multiclass imbalance problems than other imbalanced learning methods in the OVA scheme.

Table 7
The $MAvA$ (%) for different multiclass imbalanced classification methods using CART as a classifier.

	OVA	SMOTE (Chawla et al., 2002)	k-means-SMOTE (Douzas et al., 2018)	Bagging-RB (Díez-Pastor et al., 2015)	DPSE
balance scale	55.92 ± 1.2	57.09 ± 0.85	**58.74 ± 2.02**	55.82 ± 0.91	58.1 ± 0.44
car	96.67 ± 0.6	96.87 ± 0.57	97.85 ± 0.6	97.43 ± 0.56	**98.3 ± 0.24**
dermatology	**96.64 ± 0.88**	93.34 ± 0.48	93.12 ± 0.75	92.53 ± 0.41	94.18 ± 0.43
ecoli	57.99 ± 1.32	55.74 ± 1.19	50.32 ± 2.22	55.67 ± 1.61	**59.49 ± 2.32**
flare	60.87 ± 0.87	59.84 ± 0.73	**62.3 ± 1.25**	62.06 ± 1.02	62.27 ± 0.55
glass	70.95 ± 2.57	68.18 ± 2.53	70.27 ± 2.04	69.01 ± 2.32	**71.22 ± 1.73**
page-blocks	86.37 ± 0.68	86.18 ± 0.85	**94.12 ± 0.45**	86.3 ± 0.71	89.01 ± 0.32
penbased	98.05 ± 0.09	85.35 ± 0.09	91.63 ± 0.25	96.99 ± 0.05	**98.68 ± 0.05**
satimage	88.37 ± 0.29	84.17 ± 0.4	86.94 ± 0.28	86.93 ± 0.2	**89.12 ± 0.17**
shuttle	93.77 ± 0.01	93.38 ± 0.02	95.11 ± 0.01	93.31 ± 0.01	**95.75 ± 0.01**
segment	97.4 ± 0.19	96.32 ± 0	97.42 ± 0	96.61 ± 0	**97.68 ± 0.1**
vehicle	74.29 ± 0.78	68.15 ± 0.79	69.56 ± 0.68	69.41 ± 0.48	**74.67 ± 0.39**
vowel	88.98 ± 0.6	88.79 ± 0	88.78 ± 0	79.8 ± 0.64	**89.62 ± 0.46**
wine	**96.17 ± 0.8**	93.44 ± 0.83	93.35 ± 1.08	94.31 ± 0.67	94.19 ± 0.6
yeast	53.4 ± 1.06	50.95 ± 1.34	54.82 ± 1.26	**54.87 ± 0.82**	54.13 ± 0.67
zoo	92.29 ± 1.17	89.79 ± 2.78	89.3 ± 1.92	92.14 ± 1.99	**92.35 ± 1.18**
winequality-red	39.16 ± 1.51	38.44 ± 1.21	**42.79 ± 2.76**	42.28 ± 1.94	39.38 ± 0.6
winequality-white	41.01 ± 0.64	39.69 ± 0.76	42.01 ± 3.01	**43.82 ± 0.74**	43.04 ± 0.72
contraceptive	47.92 ± 0.87	48.05 ± 0.65	**50.53 ± 0.53**	48.25 ± 0.69	48.76 ± 0.6
hayes-roth	**86.95 ± 0.69**	86.15 ± 0.26	84.75 ± 0.94	85.95 ± 0.33	86.25 ± 1.2
thyroid	90.58 ± 1.89	87.81 ± 1.97	97.66 ± 1.04	**97.99 ± 1.81**	95.07 ± 0.83
newthyroid	93.77 ± 0.83	92.38 ± 1.55	93.09 ± 1.31	92.25 ± 1.2	**94.03 ± 0.55**
lymphography	79.28 ± 2.77	73.88 ± 4.11	71.79 ± 5.15	70.39 ± 3.22	**83.04 ± 0.76**
led7digit	**71.39 ± 0.7**	51.68 ± 1.79	62.1 ± 1.01	69.73 ± 0.09	70.52 ± 0.59
splice	93.74 ± 0.27	90.91 ± 0.31	91.03 ± 0.27	92.58 ± 0.17	**93.85 ± 0.2**
automobile	**82.42 ± 2.26**	73.50 ± 3.14	70.98 ± 2.19	79.98 ± 1.89	80.96 ± 0.75
cleveland	30.07 ± 1.99	29.69 ± 3.88	27.97 ± 1	29.87 ± 0.92	**31.59 ± 1.63**
Average	76.46 ± 1.02	73.32 ± 1.22	75.12 ± 1.26	75.42 ± 0.94	**77.23 ± 0.67**
Ranking($p = 2.53e{-}08$)	2.59	4.26	3.11	3.41	1.63

Table 8
The $MAvA$ (%) for different multiclass imbalanced classification methods using Random forest as a classifier.

	DES-MI (García et al., 2018)	OVO-SMB (Zhang et al., 2016)	OVO-EASY (Zhang et al., 2016)	OVO-DPSE	DPSE
balance scale	57.28 ± 0.98	59.73 ± 0.68	59.36 ± 2.38	58.08 ± 0.58	**60.35 ± 0.1**
car	96.84 ± 0.44	94.42 ± 0.67	97.41 ± 0.57	**97.91 ± 0.6**	95.64 ± 0.21
dermatology	96.85 ± 0.4	97.41 ± 0.73	93.39 ± 0.26	97.53 ± 0.22	**97.59 ± 0.34**
ecoli	71.12 ± 1.09	64.72 ± 1.58	49.54 ± 1.19	71.3 ± 0.7	**71.77 ± 0.51**
flare	60.77 ± 0.81	58.39 ± 0.75	**63.61 ± 1.09**	62.22 ± 0.51	60.99 ± 0.46
glass	76.9 ± 1.72	74.45 ± 2.29	73.83 ± 1.22	73.41 ± 0.64	**77.61 ± 0.88**
page-blocks	89.38 ± 0.82	89.09 ± 0.7	**94 ± 0.22**	92.14 ± 0.28	92.05 ± 0.25
penbased	**99.16 ± 0.03**	98.88 ± 0.03	90.75 ± 0.07	98.88 ± 0.02	99.11 ± 0.07
satimage	**90.01 ± 0.1**	89.72 ± 0.19	87.06 ± 0.07	89.92 ± 0.04	89.73 ± 0.1
shuttle	93.1 ± 0.82	95.13 ± 0.96	**95.42 ± 1.01**	93.57 ± 0.93	94.93 ± 0.74
segment	97.93 ± 0.07	97.36 ± 0	97.36 ± 0	97.36 ± 0	**98.03 ± 0.09**
vehicle	75.86 ± 0.14	75.38 ± 0.26	71.52 ± 0.25	75.89 ± 0.36	**77.3 ± 0.27**
vowel	**97.43 ± 0.24**	90.1 ± 0	90.1 ± 0	93.94 ± 0.47	96.09 ± 0.36
wine	98.72 ± 0.39	98.77 ± 0.27	92.81 ± 0.52	98.63 ± 0.33	**98.78 ± 0.79**
yeast	**58.52 ± 0.47**	52.57 ± 1.22	55.82 ± 0.35	51.99 ± 0.35	54.93 ± 0.54
zoo	89.43 ± 0	90 ± 1.28	**90.69 ± 0.7**	90 ± 0	90 ± 0.82
winequality-red	40.51 ± 1.45	41.54 ± 0.37	**44.32 ± 1.77**	41.7 ± 0.65	40.26 ± 0.41
winequality-white	43.7 ± 0.38	42.63 ± 0.19	40 ± 1.61	**44.72 ± 0.45**	44.67 ± 0.5
contraceptive	50.24 ± 0.32	50.01 ± 0.15	50.58 ± 0.61	**51.21 ± 0.43**	50.51 ± 0.13
hayes-roth	86.18 ± 0.92	84 ± 0.59	84.18 ± 0.73	85.3 ± 0.56	**87.45 ± 0.51**
thyroid	95.9 ± 1.25	90.78 ± 0.93	**97.05 ± 0.07**	95.94 ± 0.78	96.16 ± 0.37
newthyroid	96.29 ± 1.08	94.98 ± 1.02	94.73 ± 1.2	**97.89 ± 0.51**	96.54 ± 0
lymphography	**84.72 ± 1.24**	72.53 ± 2.99	69.64 ± 4.56	77.19 ± 0.3	81.47 ± 0.31
led7digit	**71.8 ± 0.42**	70.74 ± 0.37	64.86 ± 0.46	71.26 ± 0.24	70.85 ± 0.43
splice	94.7 ± 0.25	93.32 ± 0.23	94.35 ± 0.08	94.75 ± 0.11	**95.58 ± 0.03**
automobile	**80.64 ± 0.95**	70.91 ± 0.58	70.33 ± 0.59	69.97 ± 2.12	79.35 ± 0.68
cleveland	28.68 ± 2.5	28.43 ± 0.61	28.71 ± 0.65	27.13 ± 0.66	**29.87 ± 0.92**
Average	78.62 ± 0.71	76.52 ± 0.73	75.61 ± 0.82	77.77 ± 0.48	**78.8 ± 0.4**
Ranking($p = 7.77e{-}04$)	2.81	3.85	3.39	2.83	2.11

4.5. Comparison with multiclass imbalanced classification methods

In this section, we aimed to compare the DPSE with the typical multiclass imbalanced classification methods to illustrate its effectiveness. Some methods for solving multiclass imbalanced classification tasks have been proposed in the existing literature, and their effectiveness has been verified on public datasets. Therefore, three representative methods were selected from these papers as comparison methods, namely DES-MI that directly deals with multiclass imbalance problems (García et al., 2018), OVO-EASY and OVO-SMB (Zhang

Table 9
The $MAvA$ (%) for different multiclass imbalanced classification methods using SVM as a classifier.

	DES-MI (García et al., 2018)	OVO-SMB (Zhang et al., 2016)	OVO-EASY (Zhang et al., 2016)	OVO-DPSE	DPSE
balance scale	80.93 ± 0.62	82.50 ± 2.42	**86.39 ± 1.04**	80.52 ± 0.58	81.87 ± 0.39
car	97.25 ± 0.32	87.68 ± 2.06	**97.76 ± 2.33**	95.55 ± 0.67	94.52 ± 0.23
dermatology	96.86 ± 0.26	95.82 ± 2.85	93.45 ± 4.19	96.56 ± 0.44	**97.15 ± 0.1**
ecoli	64.99 ± 0.69	68.29 ± 1.74	57.35 ± 0.77	**72.16 ± 0.44**	71.89 ± 0.15
flare	**62.5 ± 0.88**	62.04 ± 2.23	62.25 ± 1.03	60.76 ± 0.46	62.35 ± 0.41
glass	**66.54 ± 1.66**	61.25 ± 2.83	62.68 ± 3.92	64.4 ± 0.49	63.49 ± 0.74
page-blocks	**92.84 ± 0.44**	72.27 ± 0.63	84.13 ± 0.54	84.91 ± 0.23	86.87 ± 0.14
penbased	99.35 ± 0.02	85.91 ± 0.69	98.74 ± 0.11	**99.41 ± 0.01**	99.37 ± 0.01
satimage	**88.51 ± 0.17**	77.95 ± 1.54	86.99 ± 0.52	88.44 ± 0.06	87.10 ± 0.05
shuttle	92.06 ± 0.24	**95.82 ± 0.19**	65.06 ± 0.26	87.65 ± 0.06	90.03 ± 0.05
segment	93.66 ± 0.26	**94.07 ± 0**	**94.07 ± 0**	**94.07 ± 0**	93.78 ± 0.07
vehicle	75.72 ± 0.54	68.79 ± 2.61	70.81 ± 0.67	75.53 ± 0.24	**78.15 ± 0.28**
vowel	94.16 ± 0.37	**95.66 ± 0**	94.04 ± 0	**95.66 ± 0**	93.59 ± 0.17
wine	97.94 ± 0.3	94.67 ± 2.86	93.69 ± 6.36	**98.26 ± 0.36**	97.52 ± 0
yeast	55.62 ± 0.48	54.8 ± 0.84	54.29 ± 1.01	56.49 ± 0.25	**57.34 ± 0.33**
zoo	88.32 ± 2.02	**92.86 ± 2.34**	87.99 ± 3.68	84.94 ± 1.72	91.75 ± 0
winequality-red	40.25 ± 1.18	38.56 ± 2.49	**41.57 ± 2.65**	36.17 ± 0.69	35.92 ± 0.38
winequality-white	41.29 ± 0.34	39.43 ± 1.02	**44.46 ± 2.52**	33.89 ± 0.44	36.77 ± 0.66
contraceptive	54.27 ± 0.28	48.26 ± 0.9	50.28 ± 0.63	54.85 ± 0.32	**56.1 ± 0.22**
hayes-roth	83.45 ± 1.06	85.07 ± 2.58	**86.59 ± 1.22**	84.73 ± 1.02	86.56 ± 0.37
thyroid	78.05 ± 1.98	85.64 ± 5.8	**97.69 ± 1.74**	70.86 ± 2.5	80.52 ± 0.05
newthyroid	**94.89 ± 0.89**	92.6 ± 2.87	93.47 ± 1.25	93.78 ± 0	94.33 ± 0.49
lymphography	64.39 ± 0.49	77.13 ± 3.96	**78.05 ± 2.49**	76.68 ± 0.72	76.73 ± 2.37
led7digit	**72.53 ± 0.62**	51.68 ± 3.17	63.23 ± 1.84	70.78 ± 0.12	71.76 ± 0.39
splice	**94.05 ± 0.16**	91.67 ± 0.39	92.63 ± 0.27	86.35 ± 0.06	85.54 ± 0.06
automobile	56.36 ± 1.59	56.91 ± 4.02	53.6 ± 1.76	55.9 ± 0.56	**59.4 ± 1.3**
cleveland	29.47 ± 1.77	30.35 ± 1.81	23.04 ± 1.65	32.19 ± 1.56	**32.27 ± 1.01**
Average	76.16 ± 0.73	73.62 ± 2.03	74.60 ± 1.65	75.24 ± 0.52	**76.39 ± 0.39**
Ranking($p = 0.16$)	2.59	3.3	2.7	3.09	2.59

et al., 2016) combined with ensemble learning and data-level sampling methods in the OVO scheme. The parameters of these methods were the same as those in the relevant literature. In addition, the proposed differential partition sampling ensemble method combined with OVO(OVO-DPSE) was also selected as the comparison method. Table 7, Tables 8 and 9 respectively correspond to the $MAvA$ obtained by CART, Random forest and SVM as the base classifier respectively on different datasets.

It can be seen from the experimental results in Tables 7–9 that the proposed DPSE is higher than the DES-MI, OVO-EASY, OVO-SMB, and OVO-DPSE in terms of the average results of all datasets. Specifically, when using CART as a classifier, the DPSE obtains the best results of the average of each class accuracy in 14 of the 27 datasets. And it has the best performance on the $MAvA$ with the lymphography dataset where the index increases 3.76%. When using Random forest as a classifier, the DPSE performs best in 10 of the 27 datasets. And it has the best performance on the $MAvA$ with the vehicle dataset where the index increases 1.41%. When SVM is used as the classifier, it has the best performance on the $MAvA$ with the automobile dataset where the index increases 2.49%. In order to further analyze the difference between each method and the DPSE on the $MAvA$, the experimental results of each method were tested by nonparametric tests. The Friedman results are shown in the last row of Tables 7–9, and results of Holm–Bonferroni test are shown in Table 10. Although the DPSE is not much different from other methods when SVM is used as a classifier, the proposed method has better performance on the average result. In other cases, there is a significant difference between the proposed method and DES-MI or OVO-EASY, OVO-SMB, OVO-DPSE when the confidence level is 95%.

5. Conclusion

Binarization is a common method to divide a multiclass classification problem into several binary problems. In this paper, the differential partition sampling ensemble in the OVA scheme is proposed to reduce the impact of data imbalance on overall classification performance.

Table 10
Results of Holm–Bonferroni test for comparing DPSE with DES-MI (García et al., 2018), OVO-SMB (Zhang et al., 2016), OVO-EASY (Zhang et al., 2016) and OVO-DPSE ($\alpha = 0.05$).

Comparison	Corrected p-value (Holm–Bonferroni)
DPSE vs. DES-MI(CART)	0.01
DPSE vs. OVO-SMB(CART)	2.0e−04
DPSE vs. OVO-EASY(CART)	0.02
DPSE vs. OVO-DPSE(CART)	5.1e−03
DPSE vs. DES-MI(Random forest)	0.26
DPSE vs. OVO-SMB(Random forest)	2.0e−04
DPSE vs. OVO-EASY(Random forest)	0.01
DPSE vs. OVO-DPSE(Random forest)	0.07
DPSE vs. DES-MI(SVM)	0.9
DPSE vs. OVO-SMB(SVM)	0.16
DPSE vs. OVO-EASY(SVM)	0.34
DPSE vs. OVO-DPSE(SVM)	0.23

The differential set of the sampling number for each binary subset is determined in advance based on the idea of incremental arithmetic progression. Then multiple balanced training sets are generated by undersampling and oversampling techniques which consider the distribution characteristics of samples according to the number of sampling, and the diversity of binary classifiers is also improved. While considering the reduction of the majority samples, it also increases the minority samples to achieve the balance between the number of positive and negative samples, thereby improving the classification accuracy.

Extensive experiments have been carried out to evaluate how well the proposed method performs on different multiclass classification problems in comparison with other typical methods. Three different classifiers CART, Random forest and SVM were used as base classifiers for experimental analysis. Experimental results show that the proposed method not only has better performance than other typical imbalanced learning methods in the OVA scheme but also has advantages compared with the imbalanced learning methods OVO-EASY, OVO-SMB, OVO-DPSE in the OVO scheme and the DES-MI, which is an ensemble method for multiclass classification tasks directly.

In order to further improve the classification accuracy, in future research, ensemble learning and dynamic selection can be considered to be embedded in the OVA scheme. In addition to considering the homogeneous method, the heterogeneous method can also be considered. By using appropriate dynamic selection methods, the advantages of different classifiers can be fully utilized to improve the classification effect.

CRediT authorship contribution statement

Xin Gao: Conceptualization, Methodology, Writing - original draft, Supervision, Funding acquisition. **Yang He:** Conceptualization, Methodology, Software development, Validation, Writing - original draft. **Mi Zhang:** Methodology, Funding acquisition. **Xinping Diao:** Software development. **Xiao Jing:** Software development, Validation. **Bing Ren:** Experimental data preprocessing. **Weijia Ji:** Experimental data preprocessing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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